

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bengaluru-560 111

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING



INTERNSHIP REPORT ON

“Monitoring and Debugging in AI-based CCTV Detection”

Submitted in partial fulfillment for the award of the degree (21INT82)

Bachelor of Engineering

In

Artificial Intelligence and Machine Learning

Submitted by:

Chandru - 1DS21AI011

Undertaken at



Paralaxiom 6B, 3rd Floor Evolve Work Studio, SNN Raj Pinnacle

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Department of Artificial Intelligence and Machine Learning,

Dayananda Sagar College of Engineering, Bengaluru

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CERTIFICATE

This is to certify that the Internship titled “**Monitoring and Debugging in AI based CCTV Detection**” carried out by **Chandru [1DS21AI011]**, a bonafide student of **Dayananda Sagar College of Engineering**, in partial fulfillment for the award of **Bachelor of Engineering in Artificial Intelligence and Machine Learning** during the academic year **2024–2025**. It is certified that all corrections/suggestions indicated have been incorporated in the report. The internship report has been approved as it satisfies the academic requirements in respect of the course Internship (21INT82).

Signature of the

Reporting Manager

Abhirup Bhattacharjee,
Manager, Data Science,
Paralaxiom

Signature of the

Internal Guide

Prof. Kavya D N,
Assistant Professor,
Dept. of AI & ML, DSCE

Signature of the

Head of the Department

Dr. Vindhya P Malagi,
Professor & Head,
Dept. of AIML, DSCE

External Viva:

Name of the Examiners

1. _____

2. _____

Signature with date

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DECLARATION

I, **Chandru**, a final year student of Artificial Intelligence and Machine Learning, Dayananda Sagar College of Engineering, Bengaluru, hereby declare that this internship report is based on the work undertaken during my ongoing internship in the role of **Intern, AI & Software Engineer Intern at Paralaxiom, Bengaluru.**

This report is submitted in partial fulfillment of the requirements for the award of the Bachelor's Degree in Artificial Intelligence and Machine Learning during the academic year 2024–2025.

Date: 16/04/2025

USN: 1DS21AI011

Place: Bengaluru

NAME: Chandru

01/Mar/2025

Subject: Internship Offer Letter

Dear Chandru,

We are glad to offer you role of an Intern in Paralaxiom Technologies Private Limited.

1. Contract

The terms governing this internship will be effective from Mar 3rd, 2025 for a period of 6 months.

2. Monthly Allowance and Benefits

Your total stipend will be INR 10,000/- per month. This will be a contractual engagement with Paralaxiom Technologies Private Limited. During the internship you can only avail unpaid leaves with prior approval from your reporting manager.

3. Evaluation during contract

This internship is subject to periodic evaluation. In case you do not meet the expected performance standards, your internship will be terminated.

4. Performance Measurement

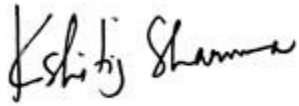
Your performance will be evaluated continuously. The extension of your internship and future opportunities within the company will depend entirely on your efficiency, dedication, consistent attendance, sincerity, good conduct, and other pertinent factors.

5. Service Rules

You will be covered by the service rules and regulations including but not restricted to the organization's norms of conduct, discipline and administrative orders and any such other rules or orders of the Company that may come in force from time to time.

We believe that this engagement will provide challenges & opportunities that will help you in your future endeavours.

Yours sincerely



Kshitij Sharma

Managing Director, Paralaxiom Technologies Private Limited.

6B, 3rd Floor Evolve Work Studio ,

SNN Raj Pinnacle Plot 7F Graphite India Rd

Whitefield Bangalore 560048

COMPLETION CERTIFICATE

ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. I take this opportunity to express my gratitude to all who have helped me to complete the Internship.

I express my sincere thanks to the Principal **Dr. B G Prasad**, for providing me adequate facilities to undertake this Internship.

I would like to thank **Dr. Vindhya P Malagi**, Head of Department – AI & ML, for providing me an opportunity to carry out Internship and for her valuable guidance and support.

I would like to thank **Abhirup Bhattacharjee**, Software Engineer, Paralaxiom, for the valuable industry insights and mentorship provided during my role as an intern in Marketing Analytics.

I would also like to extend my thanks to **Prof. Kavya D N**, internal guide, for her consistent guidance, timely feedback, and unwavering support throughout the internship. Her mentorship helped me stay aligned with both academic and industry expectations.

I would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

I would like to thank the non-teaching members of the Department of AI & ML, for helping me during the Internship.

Last but not the least, I would like to thank my parents and friends without whose constant help, the completion of Internship would have not been possible

CHANDRU [1DS21AI011]

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CHAPTER 1

INTRODUCTION

In today's data-driven world, ensuring accuracy in AI-based surveillance systems has become a critical priority—especially in the context of public safety, infrastructure security, and automated monitoring. With increasing adoption of CCTV object detection systems powered by machine learning and computer vision, the reliability of these systems directly impacts operational efficiency and security outcomes.

This internship at Paralaxiom Technologies focuses on enhancing the performance of CCTV-based object detection systems by identifying and debugging false positive alerts. False positives, which are incorrect identifications of objects or events, can lead to unnecessary alarms, resource misallocation, and reduced trust in automated surveillance solutions. The primary goal of the internship project is to build robust mechanisms to visualize, monitor, and debug false positives, ultimately contributing to the system's precision and real-world usability.

Working in close collaboration with AI engineers and mentors, the internship experience has emphasized hands-on involvement in multiple facets of the AI pipeline, including data annotation, dataset filtration, real-time visualization, and model performance monitoring. These activities are crucial for refining detection algorithms and ensuring that models generalize well across diverse environments and challenging scenarios such as low lighting, occlusions, and crowded scenes.

Visualization and monitoring tools were developed and deployed to track system behavior in real-time, enabling proactive detection of erroneous outputs and facilitating quick diagnosis of performance degradation. The intern's role involved isolating faulty detection patterns, conducting root cause analyses, and collaborating with cross-functional teams to implement corrective measures such as threshold adjustments, model retraining with curated datasets, and system parameter tuning.

Beyond technical contributions, the internship offered valuable exposure to production-grade MLOps workflows, including data ingestion pipelines, server-side data processing, and drift detection—ensuring the deployed models remain robust over time despite changing

environmental conditions. This real-world experience enhanced problem-solving skills, time management, and decision-making abilities, while providing insights into the operational challenges of maintaining AI systems at scale.

Additionally, the internship fostered effective communication skills through documentation, reporting, and presentations aimed at both technical and non-technical stakeholders, supporting knowledge sharing and collaborative progress within the organization.

By the conclusion of the internship, the project aimed to substantially reduce false alert rates and improve system reliability, thereby minimizing manual intervention and aligning with Parallaxiom's mission to deliver efficient, scalable, and real-time AI surveillance solutions that contribute meaningfully to public safety and infrastructure security.

CHAPTER 2 ABOUT THE COMPANY



Paralaxiom Technologies is an innovative AI and software engineering firm based in Bengaluru, India, known for its deep expertise in artificial intelligence, computer vision, embedded systems, and advanced data processing. The company operates at the cutting edge of smart surveillance, automation, and intelligent monitoring systems—providing robust solutions that integrate real-time insights with scalable software frameworks.

Established with a mission to drive impactful AI applications across security, transportation, industrial automation, and public infrastructure, Paralaxiom specializes in developing end-to-end products that blend hardware with intelligent software. The company’s core strength lies in its ability to deliver custom-built, AI-powered platforms that solve complex real-world challenges through continuous innovation and agile development methodologies.

Paralaxiom’s flagship offerings include smart video analytics systems, object detection frameworks, embedded AI modules, and real-time alerting platforms. These systems are designed to operate with high precision in environments like CCTV surveillance networks, industrial monitoring, and automated control systems—where quick and accurate decision-making is critical. Their solutions not only improve operational efficiency but also significantly reduce manual monitoring costs and response times.

The company maintains a strong technology stack that incorporates the latest advancements in deep learning frameworks (such as TensorFlow, PyTorch), edge computing devices, IoT integration, and cloud-native architectures. This allows Paralaxiom to deliver scalable and low-latency AI services that adapt seamlessly across diverse deployment scenarios—from edge devices with limited resources to centralized high-performance computing clusters.

Paralaxiom is heavily invested in R&D and collaboration, partnering with leading academic institutions, research labs, and government agencies to push the boundaries of applied AI. This

collaborative approach fuels innovation, enabling the company to stay ahead in emerging domains like multimodal sensor fusion, anomaly detection, and autonomous decision systems.

The company fosters a vibrant culture of experimentation, continuous learning, and cross-disciplinary teamwork. Its engineering teams leverage agile and DevOps practices to accelerate product development while maintaining high standards for quality and reliability. This agility empowers Paralaxiom to rapidly respond to evolving client needs and market trends.

Clients served by Paralaxiom range from government bodies requiring smart city surveillance solutions to industrial players seeking automated process monitoring and predictive maintenance tools. The company is committed to delivering customized solutions that align with client-specific challenges and compliance standards.

Looking forward, Paralaxiom Technologies envisions expanding its footprint globally, advancing AI ethics and explainability in its products, and driving sustainable innovation that balances technological progress with social responsibility. With a strong foundation in AI research, practical deployment expertise, and a passion for transformative impact, Paralaxiom continues to redefine the possibilities of machine intelligence across industries—empowering a future where automation enhances security, productivity, and reliability.

CHAPTER 3

SCOPE OF THE WORK

3.1 Data Annotation and Filtration

To develop a highly reliable object detection model, I meticulously curated the training dataset by applying rigorous annotation and filtration standards. I utilized advanced annotation tools (e.g., CVAT, LabelImg) to draw precise bounding boxes around objects of interest (people, vehicles, etc.), ensuring spatial accuracy and consistent labeling conventions across thousands of CCTV frames. I implemented stringent quality control measures by manually reviewing and filtering out frames exhibiting poor visibility conditions such as severe motion blur, low illumination (nighttime or shadow-heavy scenes), partial object occlusion, and ambiguous class boundaries, thereby reducing noise that could confuse the model. Additionally, I proactively flagged and catalogued complex edge cases — including densely populated scenes, adverse weather effects (rain, fog), and camera angle distortions — to ensure the training set represented a diverse range of real-world scenarios, supporting robust generalization during deployment. These efforts significantly minimized annotation bias and enhanced model robustness by fostering a balanced and high-quality training corpus.

3.2 Monitoring & Visualization of Model Predictions

Maintaining situational awareness of the deployed model's real-time performance was critical. I contributed to designing and implementing live monitoring dashboards using tools like Grafana and Plotly Dash, visualizing object detection outputs from multiple camera feeds, tracking key performance metrics such as detection counts, false positive rates, and confidence scores aggregated by time intervals and locations. I set up automated alerting systems to flag sudden spikes in false positives or drops in detection confidence, enabling rapid incident response by the operations team. Cross-validation exercises were performed by comparing model outputs against manually verified ground-truth annotations for selected samples, facilitating continuous feedback loops that guided model retraining priorities and hyperparameter tuning. This proactive monitoring framework ensured consistent detection reliability and informed data-driven operational decisions.

3.3 Debugging and Reducing False Positive Alerts

False positives posed significant operational challenges, and I played a key role in their systematic reduction by conducting forensic frame-level investigations where flagged false

positives were traced back to specific environmental triggers such as strong shadows, reflective surfaces (e.g., windows, metal), overlapping or partially visible objects, and camera motion artifacts. Collaborating closely with computer vision engineers, I developed a categorized repository of false positive examples, annotated with environmental conditions and visual characteristics, serving as a critical resource for targeted data augmentation, focused retraining, and model architecture adjustments. The impact of parameter changes was documented with before-and-after quantitative metrics and visual evidence, facilitating transparent iterative improvements. These efforts materially enhanced the trustworthiness and practical usability of the deployed detection system.

3.4 Server-Side Data Collection and Processing

Efficient and reliable server-side data handling was essential for scalable system performance. I developed and maintained robust Python automation scripts leveraging APIs and SSH protocols to ingest continuous CCTV video streams and associated detection metadata from geographically distributed edge servers. I monitored network health parameters, diagnosed frame loss, and implemented buffering and retry mechanisms to preserve frame sequencing and integrity amid network latency and packet loss issues. Additionally, I implemented data validation checks and checksum mechanisms to ensure end-to-end data fidelity, providing reliable inputs for downstream annotation and model training workflows.

3.5 Root Cause Analysis and Feedback Loop

To sustain high model quality, I systematically investigated each detection anomaly via root cause analysis (RCA). I differentiated errors originating from data issues (e.g., mislabeled samples, poor image quality) versus model limitations (e.g., weak feature representation, class confusion) through detailed manual inspection and statistical analysis. I captured comprehensive incident metadata including environmental context (time of day, weather), camera angle, detection confidence, object overlaps, and system logs to enrich RCA reports.

3.6 Data Drift Detection and Analysis

I gained early experience in post-deployment drift detection critical to model longevity by extracting statistical features from incoming data streams — including pixel intensity histograms (frame brightness), object bounding box size distributions, and scene density metrics — to quantitatively compare live input against the original training dataset baselines. I

identified drift patterns caused by evolving environmental conditions such as seasonal lighting changes, temporary obstructions from construction activity, or inadvertent camera repositioning. Documenting drift occurrences, I recommended adaptive strategies including retraining cycles with updated datasets or dataset rebalancing to re-align the model with current data characteristics.

3.7 Contribution to Server-Side Optimization and Codebase Enhancements

Collaborating with backend developers, I helped improve system throughput and robustness by refactoring server-side Python modules responsible for frame ingestion and detection logging, streamlining processing loops and eliminating redundant I/O operations to significantly reduce latency. I optimized database query patterns and file system access strategies to handle high-volume video and metadata streams without bottlenecks, utilizing indexing, caching, and batch writes. Introducing advanced error-handling logic to capture, classify, and recover from data ingestion exceptions improved system resilience and uptime.

3.8 Documentation, Reporting, and Knowledge Sharing

Clear and accessible documentation was a keystone of the project. I authored detailed technical reports on false positive case studies, server-side bug fixes, and annotation challenges, complete with visual aids such as annotated screenshots, confidence trend graphs, and confusion matrix analyses. I prepared executive summaries and visual dashboards tailored for non-technical stakeholders, translating complex model behaviors and improvements into actionable business insights. I contributed to the company's internal knowledge base by developing how-to guides, best practices, and onboarding checklists, thereby reducing future ramp-up times and facilitating smoother debugging cycles. I actively participated in team meetings and code reviews, fostering collaborative knowledge exchange and continuous process improvement. This knowledge-sharing culture enhanced overall project transparency and efficiency.

CHAPTER 4

LEARNING OBJECTIVES

The internship at Paralaxiom Technologies offered a practical and insightful experience into the field of AI deployment, specifically in the context of real-time CCTV-based object detection systems. Throughout the internship, several key learning outcomes were achieved, enhancing both technical skills and professional competencies.

4.1 Enhanced Data Annotation and Labeling Proficiency

One of the foundational outcomes was the development of precision in data annotation. By manually labeling objects in CCTV footage, I gained hands-on experience in using annotation tools to create bounding boxes and assign accurate labels. This process deepened my understanding of how high-quality, correctly labeled data contributes to the accuracy of object detection models. Additionally, I learned the importance of consistency in annotation standards, which plays a significant role in model performance during training and evaluation.

4.2 Improved Data Filtration and Quality Assurance

Another critical skill acquired was data filtration. I learned to examine video data for errors such as duplicate frames, irrelevant content, and mislabeled entries. By filtering out such data, I contributed to a cleaner and more reliable training dataset. This experience emphasized the impact of data quality on model performance and helped build a more structured approach to dataset validation.

4.3 Practical Understanding of Real-Time Model Monitoring

A significant portion of the internship involved real-time model monitoring, where I observed and analyzed the predictions made by the object detection system. This included verifying detections against expected results, identifying false positives, and logging anomalies for further review. I learned how visualization tools help in tracking model performance in dynamic environments and gained exposure to how AI systems behave in real-world deployment settings.

4.4 Strengthening Debugging and Analytical Skills

I developed a problem-solving mindset by actively participating in debugging false positive alerts. This included tracing predictions back to video frames, analyzing lighting conditions, shadow effects, and object overlaps that may lead to incorrect detections. I also explored the impact of model parameters like confidence thresholds. This systematic debugging process improved my ability to identify root causes and propose practical fixes or enhancements.

4.5 Experience in Technical Communication and Reporting

One of the essential professional skills I developed was the ability to communicate technical findings effectively. I learned to prepare concise reports, document anomalies, and summarize results using visual aids such as graphs, dashboards, and annotated frames. This helped bridge the gap between technical analysis and operational understanding, enabling smoother collaboration with other team members, including non-technical stakeholders.

4.6 Exposure to the AI Deployment Lifecycle

The internship provided an end-to-end view of how computer vision systems are prepared and deployed in production. I understood the interconnection between data collection, annotation, model inference, monitoring, and iterative improvement. This experience offered a holistic perspective on AI workflows and enhanced my ability to contribute to future projects involving real-time systems and machine learning operations.

4.7 Data Drift and Model Degradation

I gained insight into the concept of data drift—how changes in environmental conditions, camera angles, lighting, or object appearances over time can affect model accuracy. I learned to compare current prediction patterns with historical baselines to detect performance degradation. This experience helped me understand the need for regular model retraining and adaptive learning to maintain robust performance in dynamic settings.

4.8 Server-Based Data Collection Techniques

During the internship, I worked on collecting CCTV footage directly from cloud-connected servers. I learned how to interface with server APIs and manage large-scale video data retrieval through automated scripts. This included organizing data based on timestamps, camera sources, and event triggers. I also contributed to streamlining the data ingestion pipeline, ensuring that the collected footage was efficiently stored, pre-processed, and made ready for annotation and model evaluation. This experience reinforced my understanding of scalable data infrastructure and its role in real-time AI systems.

CHAPTER 5

CHALLENGES AND SOLUTIONS

5.1 Dealing with Noisy or Unreliable CCTV Data

Working with real-world CCTV footage introduced various quality issues, such as low lighting, motion blur, object occlusion, and irrelevant or cluttered scenes.

- **Challenge:** These visual inconsistencies made it difficult to annotate data accurately and affected the reliability of model evaluations. Inaccurate annotations and noisy frames contributed to false positives and poor training outcomes.
- **Solution:** A data filtration pipeline was established to remove unclear, low-quality, or irrelevant frames. Manual reviews were conducted to ensure annotation consistency, which significantly improved dataset quality and the model's training performance.

5.2 Debugging False Positive Alerts from the Object Detection Model

The model frequently generated false positives—incorrectly identifying non-existent objects in the video feed.

- **Challenge:** Pinpointing the root cause of these alerts was complex, involving multiple potential sources such as poor video quality, ambiguous scenes, or misconfigured model thresholds.
- **Solution:** A structured debugging approach was adopted. Each false alert was traced to the specific frame and examined for factors like lighting, shadows, or object overlap. Model confidence thresholds were adjusted, and the training dataset was refined based on these observations, which helped reduce false alarms.

5.3 Understanding the Real-Time Model Monitoring System

Gaining familiarity with the production-level monitoring system for real-time predictions required a learning curve.

- **Challenge:** Initially, it was difficult to interpret model output logs, identify anomalies, and understand how the monitoring dashboard reflected model behavior in live environments.

- **Solution:** With mentorship support, I gradually learned how to track prediction accuracy using visualization tools and interpret logs effectively. This helped me identify irregularities in detections and confidently contribute to real-time performance evaluations.

5.4 Managing Repetitive and Time-Intensive Tasks

Tasks like manual annotation, frame reviews, and prediction verification were time-consuming and monotonous.

- **Challenge:** These repetitive tasks led to fatigue and made it challenging to maintain productivity and focus, especially during long annotation sessions.
- **Solution:** I adopted a structured workflow using time-blocking techniques and labeling tool shortcuts. This improved both efficiency and accuracy during repetitive phases.

5.5 Handling Data Drift and Performance Degradation Over Time

Over time, the model's performance was impacted by changing environmental conditions and evolving real-world scenarios.

- **Challenge:** It was difficult to detect subtle drops in model accuracy, as there were no immediate ground truths in production, and changes were often gradual.
- **Solution:** I implemented a comparison framework to analyze prediction logs over different periods. Metrics such as time-of-day performance and region-specific accuracy were tracked. Based on these insights, recommendations for model retraining and dataset updates were made to combat data drift.

5.6 Automating and Scaling Server-Based Data Collection

Collecting and managing CCTV footage from centralized servers required careful handling due to data volume and variability.

- **Challenge:** Data retrieval was disrupted by server timeouts, inconsistent file structures, and missing timestamps, leading to gaps in the dataset.
- **Solution:** I developed automation scripts with error-checking, retry logic, and logging. Timestamps and camera metadata were used to structure and verify footage integrity. This system ensured high-quality, organized data ingestion that supported downstream annotation and training.

CHAPTER 6

ACHIEVEMENTS AND CONTRIBUTION

6.1 Enhanced Dataset Quality through Structured Annotation and Filtration

As part of the core data preparation team, I was responsible for annotating CCTV footage using bounding boxes and class labels for object detection. I applied strict quality control by only labeling clearly visible, relevant objects and following consistent annotation standards. I also manually filtered out low-quality or irrelevant frames—those with poor lighting, motion blur, or mislabeled content. These efforts reduced noise in the dataset, resulting in cleaner inputs for model training and significantly improving the model's precision and recall performance.

6.2 Identified and Reduced False Positive Alerts through Deep Debugging

A key contribution involved the analysis and reduction of false positive alerts from the deployed object detection system. I developed a methodical process for tracing these false alerts back to their source frames and identifying contributing visual factors like occlusion, reflections, or background clutter. By adjusting parameters such as detection confidence thresholds and non-max suppression values, and recommending annotation corrections, I helped the team decrease false alerts. This led to improved trust in the system's predictions during live deployment.

6.3 Built Visualization and Monitoring Dashboards for Prediction Review

To aid in real-time model evaluation, I assisted in developing and maintaining dashboards that visually represented the model's predictions on live and historical footage. These dashboards displayed bounding boxes, confidence levels, camera feeds, and detection logs, making it easier to identify anomalies like missed detections or recurring false positives.

6.4 Contributed to Workflow Optimization and Task Automation

During the annotation and review phases, I noticed repetitive manual tasks that were prone to human error. To improve efficiency, I implemented structured annotation workflows using keyboard shortcuts, batch processing techniques, and pre-labeled templates. I also contributed to developing scripts for automated file sorting and logging. These optimizations reduced task fatigue, increased daily output, and ensured consistency across multiple annotators.

6.5 Improved Real-Time Understanding of AI Model Deployment

This internship provided exposure to end-to-end AI deployment, from data collection to live inference. I observed the model's performance in real-time surveillance conditions and learned how unpredictable environmental changes—such as sudden lighting shifts, movement speed, or camera angle—could impact accuracy. Working through these real-world challenges helped me develop a practical understanding of how AI models behave beyond controlled lab environments and how to support iterative model improvements in production.

6.6 Clear and Effective Communication of Technical Findings

I regularly documented technical observations such as prediction anomalies, unusual patterns in false detections, and performance fluctuations across locations or time periods. Using visual aids like annotated screenshots, side-by-side frame comparisons, and summary tables, I communicated these findings to both technical and non-technical stakeholders. My clear and organized reporting helped the team quickly grasp the impact of specific issues and prioritize debugging or retraining activities accordingly.

6.7 Contributed to Data Drift Analysis and Mitigation

One of the more advanced contributions was detecting and analyzing data drift—subtle but significant changes in CCTV video conditions over time. I compared prediction logs across weeks and correlated performance variations with environmental changes such as lighting shifts, seasonal effects, or new objects in the scene. Based on these findings, I recommended sample collection from underperforming scenarios for retraining. This proactive approach contributed to maintaining consistent model accuracy as deployment environments evolved.

6.8 Enhanced Server-Side Data Collection Code for Robustness and Performance

I worked on backend scripts responsible for downloading video data from remote servers. The original scripts occasionally failed due to unstable connections or unstructured storage. I improved the codebase by implementing automatic retry mechanisms, integrity checks, and structured naming based on timestamps and camera IDs. These changes ensured reliable, orderly, and scalable data acquisition, which supported smoother annotation workflows and reduced delays in model updates.

CHAPTER 7

REFLECTION AND ANALYSIS

The internship experience at Paralaxiom Technologies was a transformative journey that significantly enhanced both my technical and professional skill set. It gave me a deep dive into the real-time application of artificial intelligence, particularly in the domain of CCTV-based object detection systems. One of the key takeaways was learning how crucial data quality is to the success of any AI model. Initially, I underestimated the impact of tasks like data annotation or filtration, but as the project progressed, I realized that even a few inconsistencies or noise in the dataset could result in large-scale issues such as false positives in production.

Debugging false alerts taught me to analyze scene-specific variables like lighting conditions, shadows, and object overlaps—elements that aren't covered in textbooks but make a massive difference in the real world. These tasks developed my critical thinking and problem-solving abilities, especially as I had to trace back errors to their root cause, whether it was a mislabelled frame or a thresholding issue in the model configuration.

Another major learning was the use of monitoring dashboards and visualization tools to track model behavior in real-time. This wasn't just about observing metrics—it was about identifying patterns, drawing conclusions, and making decisions on the fly. Working with such tools taught me the value of clear, real-time data representation for both technical troubleshooting and communicating insights with teammates or leads.

On the soft skills front, the internship helped me become more collaborative and organized. Whether it was documenting model behavior, participating in team reviews, or discussing debugging logs, I became more confident in presenting my work and aligning it with team goals. Exposure to a startup environment like Paralaxiom also gave me a taste of agile development cycles, tight deadlines, and cross-functional collaboration—elements that pushed me to be proactive and adaptive.

CHAPTER 8

RECOMMENDATIONS

To improve the overall quality, efficiency, and scalability of similar projects at Paralaxiom Technologies, several focused recommendations can be considered. First, the data annotation workflow could benefit greatly from the integration of semi-automated labeling tools or active learning strategies, which can intelligently prioritize and suggest labels based on model confidence. This would not only reduce the burden on human annotators but also improve consistency in labeled data. Additionally, implementing a clear set of annotation guidelines and regular quality checks can help maintain data accuracy over time.

Second, for model monitoring and visualization, the use of advanced analytics platforms or custom-built dashboards with real-time alerts and heatmaps can make it easier to identify issues such as false positives or missed detections. This can be coupled with log-based monitoring that tracks performance over various deployment scenarios—day/night conditions, weather impact, occlusions, etc.—to better understand model limitations in real environments.

Third, the debugging process can be further improved by introducing a structured error categorization system, where every false positive or anomaly is tagged with its root cause (e.g., lighting issue, mislabeling, model thresholding error). This would help in recognizing recurring patterns and guide targeted improvements. Collaboration between annotation and model development teams should also be strengthened to ensure quick feedback loops.

Fourth, from a project management perspective, setting up regular syncs and short stand-up meetings can improve team coordination, especially when working across data, engineering, and QA functions. Documenting learnings, issues encountered, and resolutions in a shared knowledge base would ensure continuity and reference value for future interns or team members.

Overall, for individual development, encouraging team members to explore related fields such as model optimization, data versioning, or even synthetic data generation could open doors for more innovative solutions in future deployments. Incorporating such proactive improvements will not only boost the performance of AI models but also strengthen the organization's capability to deploy reliable, real-world computer vision systems at scale.

CHAPTER 9

CONCLUSION

The internship at Paralaxiom Technologies has been a highly transformative experience that bridged the gap between academic learning and real-world application. Being involved in a live CCTV-based object detection project gave me valuable insight into the end-to-end lifecycle of deploying AI systems in production environments. From annotating and filtering video data to monitoring model performance and debugging false positives, each task contributed to strengthening the accuracy and reliability of the detection system. I gained firsthand experience in understanding how even small inconsistencies in data or model predictions can have significant implications in real-time scenarios, especially in surveillance and safety applications.

This experience also sharpened my technical skills in Python, data visualization, and model debugging, while encouraging me to explore deeper concepts in computer vision and AI deployment. I learned how to interpret model behavior, identify errors through visualization tools, and work collaboratively to implement solutions. In parallel, I developed stronger analytical thinking, problem-solving, and communication skills—especially when translating technical findings into actionable insights for stakeholders.

Moreover, this internship taught me the importance of teamwork, adaptability, and continuous learning in a fast-paced tech environment. Regular guidance from mentors and exposure to professional workflows helped me build confidence and maturity in handling responsibilities independently. It also sparked a deeper curiosity in building ethical, scalable, and efficient AI systems for real-world problems.

Overall, the experience not only validated my passion for AI and machine learning but also prepared me to take on more complex challenges in the field. I now feel better equipped to contribute meaningfully to future projects and research, and I look forward to applying these learnings in advanced roles and higher academic pursuits.